

Evidence-Gated Admission Control for Autonomous Quantum Research:

An Architecture and Simulation Study

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Abstract

Autonomous agents can generate and evaluate candidate quantum algorithms, but unrestricted access to quantum processing units can exhaust scarce experimental allocations before candidates have been adequately validated. We investigate how an autonomous quantum co-investigator can condition hardware access on reproducible classical evidence. We propose an evidence-gated architecture that combines independent verification, append-only provenance, an explicit resource ledger, and an Algorithmic Maturity Index measuring clean convergence, noise resilience, coherence fit, and sampling cost. We study the gate using 144 variational quantum eigensolver runs across 48 candidate-device pairs and analytical models of campaign reliability, resource allocation, and classical simulation limits. The simulator study shows that the gate exposes backend-specific failure modes and makes readiness thresholds and accuracy-cost trade-offs explicit. It also shows that threshold and backend-selection decisions depend on the problem family, device model, and operator-defined resource priorities. These findings are based on classical simulations rather than execution on accelerator clusters or physical quantum hardware. The principal contribution is therefore a falsifiable admission-control framework and evaluation protocol for autonomous quantum-information-science research.

Keywords: autonomous scientific agents; quantum information science; admission control; evidence accumulation; variational quantum algorithms; reproducibility; resource governance.

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1 Introduction

Scientific agents increasingly combine language models, callable tools, code execution, and experimental interfaces. Systems in chemistry, materials science, and algorithm discovery now plan and execute multi-step research workflows [1, 2, 3, 4]. Quantum information science (QIS) is a plausible next domain because circuit design, ansatz selection, mitigation, transpilation, and shot allocation form a large combinatorial search space. The classical simulation tier is programmable and increasingly integrated with high-performance computing (HPC) systems [16, 17, 18, 24].

The resource regime is nevertheless different from ordinary tool use. Physical quantum-processing-unit (QPU) access is usually limited within a campaign, device properties drift, and a poor result can be difficult to attribute to the ansatz, optimizer, mapping, calibration, or noise. The relevant design question is therefore not only what an agent can do, but what evidence must exist before the agent is authorized to consume hardware time.

Skill libraries remain valuable for reusable procedures [11]. A QIS agent may use skills for circuit construction, simulation, transpilation, error mitigation, or plotting. However, a skill library alone does not guarantee that evidence is persistent, that verification is independent and mandatory, or that a hardware request is refused when its evidence or budget is insufficient. These guarantees require relations among components and enforceable authority boundaries.

This paper asks:

Does evidence-gated admission control provide a more auditable and resource-aware foundation for autonomous quantum research than skill composition or ungated agent execution alone?

The paper makes five contributions:

1. It defines a requirement set for autonomous QIS campaigns and uses it to organize a capability audit of published agent and HPC-quantum systems.
2. It corrects and formalizes a campaign-reliability model in which detected failures trigger bounded retries.
3. It defines an Algorithmic Maturity Index (AMI) and separates scientific readiness from resource affordability.
4. It specifies a four-layer architecture in which the physical QPU is reachable only through a non-bypassable promotion gate.
5. It reports a simulator-based numerical instantiation and states an evaluation protocol designed to distinguish descriptive calibration from independent predictive validation.

1.1 Scope and evidence boundary

The numerical study comprises 144 real Qiskit simulations on a CPU workstation. It does not include execution on an AMD accelerator cluster or on a physical QPU. The architecture is specified but not deployed end to end. Consequently, the paper supports claims about the design, internal behavior, and falsifiability of the gate; it does not establish production reliability, hardware speedup, or QPU savings in a live campaign.

The cluster-resource request associated with this research is a separate administrative document. It is not part of the evidence reported in this manuscript and does not determine the scientific claims.

2 Related work and research gap

2.1 Autonomous scientific agents

Tool-augmented systems such as ReAct and Toolformer established a general pattern in which a language model chooses and invokes external operations [6, 7]. Scientific systems then connected planning and tool use to code, literature, and laboratories [1, 2, 3, 5]. Program-search systems provide an especially relevant precedent: FunSearch maintains candidates and accepts them only through an external evaluator [4]. The evaluator’s authority, rather than the language model’s confidence, determines acceptance.

Self-critique and iterative refinement improve some outputs, but they do not by themselves create an independent acceptance boundary [8, 9, 10]. A verifier implemented as an optional prompt or callable skill can be skipped or reinterpreted by the same agent that produced the candidate. The present work therefore distinguishes verification capability from non-bypassable verification authority.

2.2 Quantum agents and HPC-quantum integration

Quantum-agent research has addressed circuit generation, code assistance, architecture search, and laboratory automation [12, 13, 14, 15]. In parallel, HPC-quantum research has developed simulators, middleware, scheduling, and co-execution environments [18, 19, 20, 21, 22, 23]. These lines solve complementary problems. Agent research helps generate and revise candidates; middleware helps execute work on classical and quantum resources. Neither function alone specifies which candidate is sufficiently supported to consume a limited QPU allocation.

2.3 Research gap

The gap is an admission-control policy connecting agent reasoning to hardware execution. Such a policy needs at least four properties:

- evidence persists outside the conversational transcript;
- an independent process can reject a claim;
- scientific readiness is expressed by an auditable decision rule; and
- a resource ledger can refuse an otherwise mature request that exceeds the remaining allocation.

The contribution is not the claim that skills cannot write records or call verifiers. They can. The narrower claim is that reusable capabilities do not, by themselves, enforce persistence, independence, or refusal.

3 Research questions and evaluation logic

We organize the work around four research questions.

- RQ1.** Which capabilities are missing when existing scientific-agent and HPC-quantum architectures are evaluated against the needs of a scarce-resource QIS campaign?
- RQ2.** Under an explicit error-and-retry model, how does mandatory external verification change the reachable length of a dependent campaign?

- RQ3.** Can a multi-axis readiness index reveal why a candidate fails, while exposing the threshold and cost assumptions that govern promotion?
- RQ4.** What evidence would be required to demonstrate that the gate predicts useful hardware outcomes rather than merely reproducing quantities already encoded in the index?

RQ1 and RQ2 are addressed analytically. RQ3 is addressed descriptively on the available simulator grid. RQ4 defines the independent validation still required. This separation prevents simulator calibration from being presented as physical-hardware validation.

4 Capability audit

4.1 Coding framework

The source study assembled 93 references and selected 23 systems making architectural claims about autonomous or semi-autonomous scientific work. Each system was coded on a three-level scale: 0 for absent, 1 for partial or episodic behavior, and 2 for a named first-class mechanism. Table 1 defines the nine capabilities.

Table 1: Capabilities required for an evidence-gated QIS campaign.

Code	Capability	Operational definition
C1	Persistent hypothesis state	Versioned candidates, parameters, lineage, and status survive across sessions.
C2	External verifier	A process can reject the primary agent’s output, and the agent cannot override the verdict.
C3	Campaign planning	An explicit task graph tracks progress, dependencies, and bounded replanning.
C4	Scarce-resource governance	A non-replenishing campaign budget is attached to admission decisions.
C5	Scheduler integration	The system submits jobs to and reads results from a batch scheduler.
C6	Hardware-aware compilation	Device topology, basis, timing, and mapping affect evaluation.
C7	Quantitative stopping rule	A numeric criterion and threshold determine continuation or promotion.
C8	Machine-readable provenance	Code, environment, seeds, inputs, outputs, and decisions are replayable.
C9	Domain-grounded cost model	Costs are represented in shots, QPU-seconds, and accelerator-hours.

The reported audit found that no system covered all nine capabilities, even partially. External verification and provenance appeared comparatively often, while scarce-resource governance and quantitative stopping were not represented as first-class mechanisms. Program-search systems were strongest on persistent candidates and external evaluation; HPC-quantum middleware was strongest on scheduling, compilation, and cost. The result is best interpreted as complementarity, not as a ranking of systems built for different purposes.

4.2 Audit limitations and required strengthening

The audit is an authored coding exercise, not yet a systematic review with independent raters. Before external publication, the accompanying corpus should include search strings, search dates, inclusion and exclusion criteria, evidence excerpts, two independent coders, adjudication records, and an inter-rater agreement statistic. Claims such as “none” should be restricted to the coded corpus and date.

5 Why enforced verification changes campaign reliability

5.1 A corrected bounded-retry model

Consider a campaign of H dependent steps. A fresh attempt succeeds with probability p . If it fails, an external verifier detects the failure with recall ρ . A detected failure triggers a new attempt, with at most k retries after the initial attempt. A missed failure is propagated and causes the dependent campaign to fail.

Let

$$a = (1 - p)\rho. \quad (1)$$

The probability that one step ultimately succeeds is the probability of success immediately or after one of the detected failures:

$$p_{\text{eff}} = p \sum_{j=0}^k a^j = p \frac{1 - a^{k+1}}{1 - a}. \quad (2)$$

The end-to-end success probability is

$$P_{\text{campaign}}(H) = p_{\text{eff}}^H, \quad H_{50} = \frac{\log 0.5}{\log p_{\text{eff}}}. \quad (3)$$

For a skill-only policy with no independent rejection, $p_{\text{eff}} = p$. At $p = 0.95$, this gives $H_{50} = 13.5$ dependent steps. A verifier with $\rho = 0.70$ and $k = 2$ yields $p_{\text{eff}} = 0.984414$ and $H_{50} = 44.1$. A stronger verifier with $\rho = 0.90$ and $k = 3$ yields $p_{\text{eff}} = 0.994760$ and $H_{50} = 131.9$, approximately a 9.8-fold increase over the ungated baseline.

Table 2: Analytical campaign horizon under the corrected retry model. Values are model outputs, not deployed-agent measurements.

Policy	p	ρ	Retries k	H_{50}
No external verification	0.95	—	0	13.5
Weak verifier	0.95	0.70	2	44.1
Strong verifier	0.95	0.90	3	131.9

This model corrects a previous formulation that multiplied the base error probability by repeated verifier-miss probabilities. That formulation did not match a retry process and produced values inconsistent with the reported horizons. Equation (2) explicitly models fresh attempts following detected failures.

5.2 Interpretation and limitations

The analysis isolates an architectural effect: mandatory rejection introduces ρ and k , while a better skill primarily changes p . It does not demonstrate that real errors are independent, that verifier recall is constant, or that a fresh attempt is identically distributed. Correlated errors and shared proposer-verifier blind spots will reduce the gain. Therefore, the analytical horizon is a falsifiable prediction and likely an upper bound, not an empirical performance claim.

6 Algorithmic Maturity Index

6.1 Purpose and design requirements

The AMI is an admission score, not a universal measure of algorithm quality. It asks whether the available classical evidence is sufficient to justify a hardware experiment on backend b . Its requirements are:

- R1. bounded, interpretable axes;
- R2. non-compensatory aggregation;
- R3. device-relative evaluation;
- R4. estimation without using the hardware access being governed; and
- R5. replication with uncertainty reported explicitly.

Let E_0 be the exact or best-known reference energy, E_{clean} the noiseless variational minimum, E_{noisy} the result under a calibrated device model, T the critical-path circuit duration, T_2 the dephasing time, F a fidelity proxy, $\|H\|_1$ the Hamiltonian 1-norm, ϵ the useful energy resolution, and B the QPU-time budget per candidate. Define

$$\text{sat}(x; \ell, h) = \text{clip}\left(\frac{x - \ell}{h - \ell}, 0, 1\right). \quad (4)$$

The four axes are

$$A_1 = \text{sat}\left(-\log_{10} \frac{|E_{\text{clean}} - E_0|}{\epsilon}; -1, 0\right), \quad (5)$$

$$A_2 = \text{sat}\left(-\log_{10} \frac{|E_{\text{noisy}} - E_{\text{clean}}|}{\epsilon}; -1.5, 0\right), \quad (6)$$

$$A_3 = \text{sat}\left(-\log_{10} \frac{T}{T_2}; -0.5, 1.5\right), \quad (7)$$

$$A_4 = \text{sat}\left(-\log_{10} \frac{t_{\text{QPU}}}{B}; -1.5, 0\right), \quad (8)$$

where

$$N = \frac{(\|H\|_1 / \sigma_{\text{target}})^2}{F^2}, \quad t_{\text{QPU}} = NT. \quad (9)$$

These are planning proxies. In particular, the shot expression does not replace term grouping, covariance-aware estimators, or empirical variance measurement. A deployment should substitute a measurement-cost estimator appropriate to the Hamiltonian and protocol.

6.2 Aggregation and promotion

The axes are combined by a weighted geometric mean,

$$\text{AMI}_b = \prod_{i=1}^4 A_{i,b}^{w_i}, \quad \mathbf{w} = (0.30, 0.30, 0.25, 0.15). \quad (10)$$

The geometric form is non-compensatory: an axis approaching zero pulls the total score toward zero. The weights are policy choices, not physical constants, and must be included in sensitivity analysis.

For replications $X_{b,1}, \dots, X_{b,r}$ of the candidate’s AMI, define a one-sided lower confidence bound $L_{1-\alpha}(\mu_b)$. Promotion occurs when

$$L_{1-\alpha}(\mu_b) \geq \theta. \quad (11)$$

This statement replaces the ambiguous expression $\Pr[\text{AMI}_b \geq \theta] \geq 1 - \alpha$: a confidence bound on a mean and a probability of exceeding a threshold are different estimands. With only three replications, a parametric confidence bound is fragile. The simulator study therefore reports means and dispersion descriptively; an operational gate should preregister the interval estimator and use enough independent replications to support it.

7 Evidence-gated architecture

The architecture has four layers (Figure 1). Authority flows downward only through defined interfaces, while evidence flows upward from execution to governance.

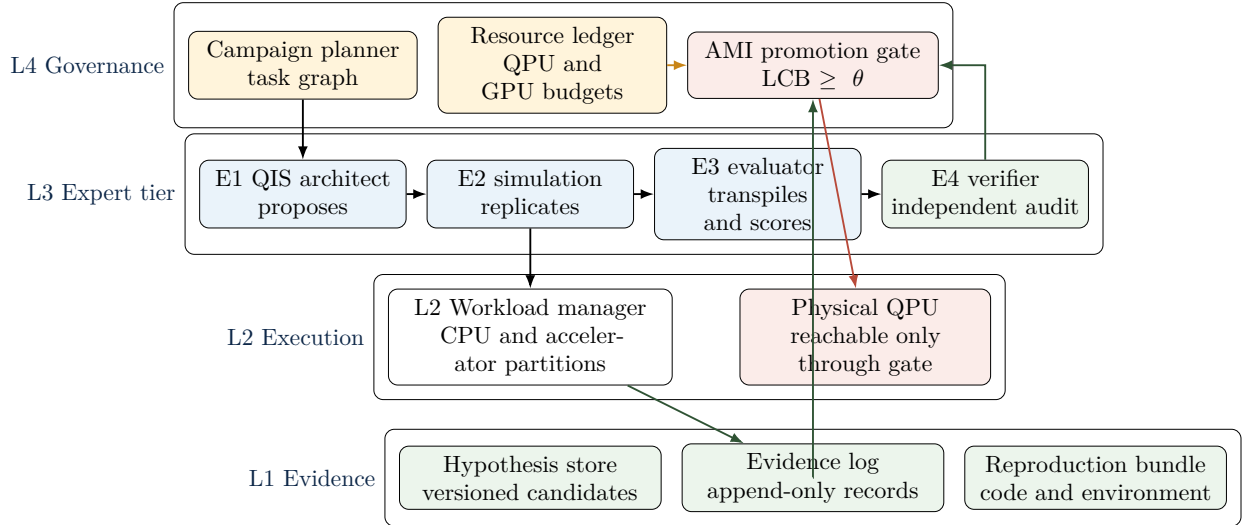


Figure 1: Four-layer evidence-gated co-investigator. The gate reads replicated evidence and the resource ledger; no expert has a direct execution path to the physical QPU.

7.1 L1: evidence and provenance

The hypothesis store treats each candidate as a versioned object rather than a passage of transcript. The evidence log is append-only and records seeds, code version, environment, device model, metrics, uncertainty, and cost. Rejected candidates remain available as negative results. The reproduction bundle connects each manuscript claim to an executable workflow and machine-readable output [26].

7.2 L2: execution substrate

The execution layer exposes CPU, accelerator, and QPU resources through a workload manager such as Slurm [25]. Simulation and transpilation run before hardware admission. The QPU may be integrated through middleware or an external service, but its authorization endpoint accepts requests only from the gate.

7.3 L3: routed experts

The expert tier separates proposal, simulation, evaluation, and adversarial verification. This separation is an authority boundary rather than merely a collection of prompts. The verifier runs on an unconditional path and writes a verdict that the proposer cannot suppress. Separate processes, models, tools, and evidence views should be used where practical to reduce shared failure modes.

7.4 L4: campaign governance

Governance owns the task graph, the AMI policy, and the resource ledger. The gate judges whether the evidence supports a hardware experiment; the ledger judges whether the campaign can afford it. A request can therefore be rejected for scientific immaturity, insufficient uncertainty control, or budget exhaustion, with each reason logged separately.

8 Numerical instantiation

8.1 Experimental setup

The reported simulator grid studies the transverse-field Ising Hamiltonian, for which an exact ground-state energy is available at the tested sizes. Candidate circuits are **EfficientSU2** ansatzes with

$$n \in \{4, 6, 8\}, \quad d \in \{1, 2, 3, 4\}, \quad e \in \{\text{linear}, \text{full}\}. \quad (12)$$

These choices yield 24 circuit configurations. Each is evaluated against stylized superconducting and trapped-ion device models, giving 48 candidate-device pairs. Three independent optimization seeds produce 144 variational runs. Noiseless optimization uses a statevector simulator; noisy evaluation uses density-matrix propagation. The full grid reportedly required 46 minutes on ten CPU cores.

Table 3: Simulator-study design.

Element	Specification
Problem family	Transverse-field Ising ground-state estimation
Ansatz	EfficientSU2
Qubits	4, 6, and 8
Repetition depth	1, 2, 3, and 4
Entanglement	Linear and full
Device models	Stylized heavy-hex superconducting and all-to-all trapped ion
Replications	Three independent seeds per candidate-device pair
Total	48 pairs and 144 variational runs
Hardware evidence	CPU simulation only; no accelerator or physical-QPU execution

8.2 Descriptive findings

The reported grid reveals backend-specific binding axes. The superconducting model is principally limited by noise resilience, whereas the trapped-ion model is principally limited by sampling time. This is the most useful function of the multi-axis representation: it identifies a repair direction that a scalar readiness value alone would obscure.

At the originally proposed threshold $\theta = 0.90$, eight pairs are promoted. At $\theta = 0.80$, sixteen pairs are promoted. When success is defined as noisy simulated energy within 5% of the exact value, the reported descriptive precision is 1.00 at both thresholds, while recall increases from 0.40 to 0.80. The cost share rises from 5.8% to 25.3% of the simulated QPU-seconds assigned to running all candidates.

These values are useful for internal calibration but are not independent predictive validation. The success label depends on $E_{\text{noisy}} - E_0$, while A_1 and A_2 encode $E_{\text{clean}} - E_0$ and $E_{\text{noisy}} - E_{\text{clean}}$. The target is therefore algebraically related to components of the predictor. In addition, the same grid is used to inspect and select the threshold. The result demonstrates that the implemented score is internally coherent; it does not demonstrate that AMI predicts unseen hardware outcomes.

8.3 Backend choice as an explicit trade-off

The simulator grid places the two device families at different points on an accuracy-time trade-off. The trapped-ion model has lower simulated energy error but much greater circuit time. Define

$$J_b(\lambda) = \log_{10}(\text{relerr}_b) + \lambda \log_{10}(t_{\text{QPU},b}), \quad (13)$$

where λ expresses the operator’s exchange rate between accuracy and QPU time. The reported majority backend changes near $\lambda \approx 0.43$. This value is specific to the grid; the transferable conclusion is that “optimal backend” is conditional on an explicit resource preference, not intrinsic to an algorithm.

9 Independent validation protocol

The next study should separate calibration from evaluation and implement actual agent baselines. Table 4 defines the minimum protocol.

Table 4: Required validation beyond the current simulator calibration.

Component	Minimum design	Primary outcome
Calibration split	Select weights and θ using designated Hamiltonians and seeds only	Fixed preregistered gate
Held-out split	New Hamiltonians, ansatze, qubit counts, and device calibrations	Precision, recall, calibration error
Skill-only baseline	Implemented agent with tools but no enforced evidence store or gate	Productive runs per fixed budget
Evidence-only ablation	Persistent evidence but unconditional hardware promotion	Value of persistence alone
Gate-only ablation	AMI gate without independent verifier	Value of verifier authority
Full architecture	Evidence, verifier, gate, and ledger	End-to-end resource efficiency
Hardware outcome	Physical QPU result or preregistered hardware proxy not used in AMI	Independent validity
Statistical plan	Confidence intervals across problem instances and campaigns	Uncertainty and generalization

The primary endpoint should be productive hardware outcomes per fixed QPU budget, accompanied by promotion precision, promotion recall, QPU-seconds, accelerator-hours, and the number of candidates abandoned with useful diagnostic evidence. Campaign horizon should be measured from actual dependent task graphs rather than inferred only from Equation (3).

10 Illustrative HPC deployment constraints

The original study included an analytical four-MI300X reference system with 768 GB aggregate HBM and a 100 Gbps inter-node link. For complex-double statevectors, the memory model

$$M_{SV}(n) = 16 \cdot 2^n \text{ bytes} \quad (14)$$

gives an aggregate upper bound of 35 qubits, while a dense density matrix

$$M_{DM}(n) = 16 \cdot 4^n \text{ bytes} \quad (15)$$

gives 17 qubits. These values are capacity bounds for that illustrative configuration, not measurements and not specifications for the current AMD University Program cluster.

The operational lesson is broader than the numbers. Distributed state sharding can be communication-bound, whereas independent candidates and replications are embarrassingly parallel. The admission gate already requires replications, so a deployment should prefer candidate- and seed-parallel execution before distributed state sharding. Actual memory ceilings, collective behavior, and simulator support must be measured on the awarded system.

The cluster request, node-hour estimates, application schedule, and AMD-specific milestones remain outside this paper. A future deployment paper may report them after execution, with configuration, job scripts, profiler traces, and numerical-parity checks.

11 Threats to validity

No independent hardware outcome. The current success label is derived from simulated energies related to AMI inputs. It cannot establish predictive validity on physical hardware.

Narrow problem family. Small transverse-field Ising instances with hardware-efficient ansatze do not represent chemistry Hamiltonians, adaptive ansatze, barren plateaus, or strongly entangling circuits.

Small replication count. Three seeds describe dispersion but provide weak support for tail probabilities or confidence-bound decisions. Operational replication counts must be tied to a preregistered error target.

Stylized device models. The superconducting and trapped-ion parameters represent families, not named devices with time-stamped calibration data. Numerical thresholds and backend inversions require recalibration.

Authored capability coding. The capability audit has not yet reported independent coding or inter-rater agreement. The corpus may under-code systems whose papers omit implementation details.

Analytical campaign assumptions. Equation (2) assumes independent fresh attempts and fixed verifier recall. Correlated failures and shared blind spots reduce the predicted horizon.

Architecture not deployed. The paper specifies authority boundaries and data flows but does not demonstrate multi-agent coordination, scheduler integration, or non-bypassable enforcement in production.

12 Reproducibility requirements

An external release should not state that artifacts are available until a permanent archive exists. The release package should contain:

- the literature corpus, search protocol, coding evidence, and adjudication records;
- the AMI implementation and unit tests;
- raw per-run metrics and candidate-level decisions;
- threshold and weight sensitivity analyses;
- scripts reproducing every table and figure;
- exact package versions, container or lock file, and random seeds;
- a tagged repository release and archival DOI; and
- a claim-to-artifact index linking manuscript statements to outputs.

Until these materials are deposited, numerical findings should be described as reported results from the available study rather than independently reproducible results.

13 Discussion

The central design insight is an authority distinction. Skills extend what an agent can do; admission control determines what the agent is permitted to do under evidence and budget constraints. A skill can create evidence, call a verifier, or estimate cost, but only an enforced architecture can require those steps and prevent a direct path to hardware.

The AMI is useful chiefly because it makes assumptions inspectable. Its axes expose why a candidate fails; its weights reveal policy preferences; its threshold exposes a precision-recall trade-off; and its confidence procedure determines how instability is treated. The index should not be defended as universally correct. It should be treated as a measurable policy that can be calibrated, challenged, and replaced.

The simulator study also illustrates the difference between descriptive and predictive evaluation. Reconstructing a simulated success label from related energy quantities can verify implementation consistency, but it cannot establish generalization. Independent outcomes, held-out problems, and ablations are therefore not optional extensions; they are the evidence needed for the paper’s strongest claim.

The architecture may transfer to other scarce experimental regimes, including beam time, cryogenic microscopy, clinical samples, or destructive testing. The transferable component is not the quantum-specific AMI. It is the pattern of persistent evidence, independent verification, explicit cost, and non-bypassable admission.

14 Conclusion

This paper proposes evidence-gated admission control for autonomous quantum research. The architecture combines versioned hypotheses, append-only evidence, independent verification, a multi-axis readiness index, and a resource ledger that can refuse unaffordable requests. A corrected bounded-retry model predicts that mandatory verification can substantially extend a dependent campaign under its stated assumptions. A 144-run simulator study demonstrates how the AMI exposes backend-specific limitations and operator-defined accuracy-cost trade-offs.

The present evidence does not yet show that the gate predicts physical-QPU success or that the full co-investigator operates reliably on an accelerator cluster. The threshold analysis is descriptive, the capability audit requires independent coding, and the artifact package requires a permanent release. These limitations narrow but strengthen the contribution: a falsifiable architecture and evaluation protocol for deciding when an autonomous system has accumulated enough evidence to request scarce quantum hardware.

Data and artifact status

The manuscript is accompanied by an editable LaTeX source. The empirical and audit artifacts named in the study must be deposited in a versioned public archive before a claim of full reproducibility or artifact availability is made.

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